

Paper 073 — The Diffusion Coarse-Graining Theorem (DCGT)

A Generative Paper of the Event Density Program

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Preamble: What This Paper Does NOT Claim

1. **The 13 substrate primitives are not derived** (Paper_087).
2. **DCGT is not the only possible substrate→continuum bridge.** Other coarse-graining procedures (Wilsonian RG, hydrodynamic projection, mode-coupling) may apply in specific regimes; DCGT is the one substrate-level bridge derived from substrate-graph diffusion of V1/V5 content.
3. **DCGT does not derive continuum-level coupling constants or transport coefficients.** It supplies the *form* of continuum equations (diffusion-type, propagator-type, constitutive-law-type); the *coefficients* (diffusion constant, viscosity, conductivity, etc.) are inherited from value-layer empirical content and substrate-level matching at the coarse-graining scale.
4. **DCGT is regime-conditional.** It applies in the hydrodynamic-window scale separation $\ell_{\text{ED}} \ll R_{\text{cg}} \ll L_{\text{flow}}$. Outside this window (strong-gradient regimes, near-substrate-scale physics, near-singularity regimes), DCGT breaks down.
5. **No claim that DCGT derives standard QFT renormalization.** DCGT operates one structural level below RG: it produces the continuum theory whose RG flow is then studied by standard techniques.
6. **DCGT is not a derivation of Einstein equations.** The substrate→continuum bridge produces flat-spacetime continuum equations under the standard hydrodynamic window; curvature emergence is the subject of the Arc ED-10 program, not DCGT.
7. **No claim that DCGT is exhaustive across all ED arcs.** Soft-matter Maxwell relaxation, gravity-arc cumulative-strain, BH-arc Hawking spectrum, and Q-COMPUTE multiplicity-cap behavior all use DCGT differently; the regime-specific applications are developed in arc-specific papers.

Abstract

The **Diffusion Coarse-Graining Theorem (DCGT)** is the substrate→continuum bridge in the Event Density Generative Substrate Ontology: given the 13 postulated primitives (Paper_087) + V1 inheritance (Paper_089) + V5 dependence (Paper_090), substrate-level kernel-mediated content coarse-grains under the **hydrodynamic-window scale separation** $\ell_{\text{ED}} \ll R_{\text{cg}} \ll L_{\text{flow}}$ to continuum diffusion-type / propagator-type / constitutive-law equations. The theorem has three load-bearing components: (1) **scale separation** — substrate-graph operations at scale ℓ_{ED} produce well-defined coarse-grained content at scale R_{cg} that depends on macroscopic flows at scale L_{flow} but is decoupled from substrate microstructure; (2) **diffusion-form emergence** — V1's finite-width retarded envelope coarse-grains to a continuum diffusion operator under standard substrate-graph averaging; (3) **cross-domain instantiations** — DCGT produces continuum Maxwell relaxation in soft-matter (V5 soft-regime), retarded / Feynman / advanced propagator

hierarchy in QFT (V5 QFT-regime), Lieb–Robinson velocity bound in condensed matter, and curved-spacetime entanglement at horizons (V5 BH-regime). **Verdict per Paper_095: M3 (form-IDENTIFIED + value-INHERITED)** within the hydrodynamic-window scale-separation regime — diffusion-form emergence is substrate-derived (form); transport coefficients are inherited via empirical matching (value); regime is explicit $A \rightarrow$ regime scale-separation. The empirical case rests on cross-domain reach: DCGT is load-bearing for Paper_090 (V5 cross-chain kernel applications), the gravity arc (Papers_027/029/030/031), the BH arc (Papers_039/047), the soft-matter arc, and the entire Wave-3 corpus.

1 Introduction

Standard physics operates in the continuum (fields, propagators, diffusion equations, hydrodynamics); ED’s substrate ontology operates on discrete substrate-graph content (chains, channels, polarity-transport, commitment). **DCGT is the bridge** — substrate-level kernel-mediated content coarse-grains under the hydrodynamic-window scale separation $\ell_{ED} \ll R_{cg} \ll L_{flow}$ to continuum diffusion-type / propagator-type / constitutive-law equations.

The paper (i) states DCGT formally with the regime assumption explicit; (ii) derives diffusion-form emergence from V1 finite-width + substrate-graph averaging; (iii) catalogues cross-domain instantiations (Maxwell relaxation, QFT propagators, Lieb–Robinson velocity, KMS-thermal); (iv) specifies breakdown conditions.

Corpus role. DCGT is upstream of Paper_090 (V5 cross-domain), Paper_027 (Newton’s G), Paper_039 + Paper_047 (BH arc), and is inherited by the entire Wave-3 corpus as the substrate $\rightarrow$ continuum bridge at M3-equivalent inheritance level — every continuum-level empirical prediction in the corpus traces structurally through DCGT.

2 Primitive Inputs

This paper takes the following ED substrate primitives as **postulated** (Paper_087):

- **P02 (Participation).**
- **P03 (Channel + locus indexing; spatial homogeneity).** Translation-invariance is load-bearing for the substrate-graph averaging.
- **P04 (Bandwidth as non-negative additive scalar).** Additivity is load-bearing for coarse-grained substrate content.
- **P05 (Polarity-transport along edges).**
- **P06 (Spatial dimension $D = 3 + 1$).** Continuum diffusion operators are D -dimensional.
- **P07 (Channel structure).**
- **P08 (Substrate scale ℓ_{ED}).** Defines the substrate-graph granularity below which DCGT does not resolve.
- **P09 ($U(1)$ -valued polarity).**
- **P10 (Rule-type primitive).**
- **P11 (Commitment irreversibility).**
- **P13 (Time homogeneity).**

V1 inheritance (Paper_089): finite-width retarded structure; substrate-level kernel that coarse-grains to continuum diffusion / propagator operators.

V5 dependence (Paper_090): cross-chain correlation kernel; coarse-grains to continuum constitutive laws (Maxwell relaxation, Lieb–Robinson bound, KMS-thermal structure).

Empirical / value-layer inputs: Substrate scale ℓ_{ED} empirically identified with ℓ_P (Paper_027 §5.3); coarse-graining scale R_{cg} regime-dependent; macroscopic flow scale L_{flow} regime-dependent. No primitive forcing is invoked.

3 Load-Bearing Step Audit

Step	Status	Source / justification
13 primitives postulated	P	Paper_087
V1 finite-width retarded structure	D	Paper_089
V5 cross-chain correlation kernel	D	Paper_090
Hydrodynamic-window scale separation	A→regime	Regime assumption , §4.1
Substrate-graph averaging at R_{cg} produces well-defined continuum content	D	§4.2 from P03 + P04
V1 envelope coarse-grains to diffusion operator	D	§5 from V1 finite-width + averaging
V5 envelope coarse-grains to memory-kernel diffusion	D	§6 from V5 finite-memory + cross-chain averaging
Continuum Maxwell relaxation form (exponential decay)	D	§7.1 from V5 Markovian-limit
Continuum Maxwell relaxation numerical τ_{Maxwell} value	I	§7.1. Empirical matching.
Continuum propagator hierarchy (QFT) form	D	§7.2 from V1/V5 coarse-graining
Lieb–Robinson velocity bound form $v_{LR} \sim \ell_{V5}/\tau_{V5}^{\text{ent}}$	A→position	§7.3. Identification by dimensional analogy.
KMS-thermal structure at horizons (BH)	D	§7.4 from V5 imaginary-time + DCGT
Breakdown taxonomy	P	§8.1–§8.4
Numerical transport coefficients	I	Empirical inheritance
Verdict M3 (form-IDENTIFIED + value-INHERITED) within hydrodynamic-window regime	A→position	Per Paper_095 §3.3

One row is **A→regime** (hydrodynamic-window scale separation). One row is **A→position** for the Lieb–Robinson identification. One row is **A→position** for the verdict-tier. All other substantive rows are P, D, or I.

4 Setup — Scale Separation and Substrate-Graph Averaging

4.1 The hydrodynamic-window scale separation (regime assumption)

DCGT operates in the **hydrodynamic-window scale separation**:

$$\ell_{\text{ED}} \ll R_{\text{cg}} \ll L_{\text{flow}},$$

where ℓ_{ED} is the substrate scale (P08), R_{cg} is the coarse-graining scale (the spatial / temporal extent over which substrate-graph content is averaged to produce continuum-level quantities), and L_{flow} is the macroscopic flow scale (the spatial / temporal extent over which continuum-level quantities vary appreciably).

This is a regime assumption, not a derivation. DCGT’s conclusions apply when the scale separation holds. Outside this window, DCGT breaks down (§8).

In practice, R_{cg} is chosen anywhere in the scale-separated window; coarse-grained quantities are insensitive to the specific choice within the window (the hallmark of effective-theory regime stability).

4.2 Substrate-graph averaging

A **substrate-graph average** of a substrate-level quantity $\mathcal{Q}(u, t)$ at coarse-graining scale R_{cg} is:

$$\langle \mathcal{Q} \rangle_{R_{\text{cg}}}(\bar{u}, \bar{t}) = \frac{1}{V_{R_{\text{cg}}}} \int_{|u-\bar{u}| < R_{\text{cg}}} \int_{|t-\bar{t}| < R_{\text{cg}}/c} \mathcal{Q}(u, t) d\mu_{\text{substrate-graph}}$$

where $V_{R_{\text{cg}}}$ is the substrate-graph volume of the coarse-graining cell and $d\mu_{\text{substrate-graph}}$ is the substrate-graph measure (substrate channels per substrate cell).

Well-definedness. By P03 (spatial homogeneity), translation-invariance of substrate-graph operations means the average depends only on $(\bar{u}, \bar{t})$ — the coarse-graining center — not on the specific substrate channels enumerated within the cell. By P04 (bandwidth additivity), substrate bandwidth contributions sum additively within the cell.

4.3 What survives coarse-graining

Substrate-level content that varies on scales $\sim \ell_{\text{ED}}$ is **averaged out**: substrate-graph microstructure (specific channel labels, specific commitment-event timings) becomes invisible at scale R_{cg} .

Substrate-level content that varies on scales $\sim L_{\text{flow}}$ **survives**: macroscopic flows of bandwidth, polarity, strain content are captured by the coarse-graining.

5 V1 Kernel $\rightarrow$ Continuum Diffusion Operator

5.1 V1 envelope at substrate-graph scale

By Paper_089, V1 has the canonical form:

$$K_{V1}(x, x') = \theta(t - t') G(\sigma(x, x')/\ell_{\text{ED}}^2).$$

The envelope G is bounded, finite-width with characteristic scale ℓ_{ED} , decaying outside the substrate-graph scale.

5.2 Coarse-graining V1 to a diffusion-type operator

Coarse-grain V1 at scale R_{cg} via the substrate-graph averaging of §4.2. For separations $|x - x'| \gg \ell_{\text{ED}}$, the V1 envelope becomes effectively local at scale R_{cg} — the kernel acts like a **point-supported operator** with effective range $\ell_{\text{ED}} \ll R_{\text{cg}}$.

The continuum-limit operator $\hat{D}_{\text{cont}}$ defined by

$$\hat{D}_{\text{cont}} \psi(x) = \int K_{V1}(x, x') \psi(x') d^4x'$$

(restricted to retarded support) reduces in the hydrodynamic window to a **local differential operator** acting on the coarse-grained field $\psi(\bar{x})$. The standard expansion (Taylor in $(x - x')/R_{\text{cg}}$ to second order) gives:

$$\hat{D}_{\text{cont}} \psi(\bar{x}) \approx \left[a_0 + a_1 \partial_t + D \nabla^2 + \mathcal{O}(\partial^3) \right] \psi(\bar{x})$$

where a_0 is the zeroth-moment integral of the V1 envelope, a_1 is the first-moment in time, and $D = (1/6) \int G |\Delta \vec{x}|^2 d^4(\Delta x)$ is the second-moment in space (giving the diffusion constant).

5.3 Diffusion-form result

In the hydrodynamic window, V1's coarse-grained action on substrate content reduces to a **diffusion-type continuum operator**:

$$\hat{D}_{\text{cont}} \approx \partial_t - D \nabla^2 + (\text{higher-order corrections})$$

(after suitable normalization absorbing the zeroth-moment a_0). This is the substrate→continuum diffusion form: V1 cross-chain content propagates at the substrate level via finite-width retarded kernel, and coarse-grains to a continuum diffusion equation with diffusion constant D set by V1's second-moment substrate-graph structure.

5.4 Where the diffusion constant comes from

The numerical value of D is set by V1's specific functional shape (envelope details) and the substrate scale ℓ_{ED} . For a generic V1 envelope $G \sim e^{-\sigma/\ell_{\text{ED}}^2}$, $D \sim \ell_{\text{ED}}^2/\tau_{V1}$ where $\tau_{V1} \sim \ell_{\text{ED}}/c$. This gives $D \sim c\ell_{\text{ED}}$ at the substrate scale.

In specific regimes, the effective D is renormalized by substrate-coefficient matching to empirical values — **value-layer empirical content**, not substrate-derived. DCGT supplies the *form*; the *coefficient* is matched.

6 V5 Kernel → Continuum Memory-Kernel Diffusion

6.1 V5 envelope

By Paper_090, V5 has the canonical form:

$$K_{V5}(u_A, t_A; u_B, t_B) = \theta(t_A - t_B) F_{V5}(\sigma/\ell_{V5}^2, (t_A - t_B)/\tau_{V5}).$$

V5 has both finite spatial extent (ℓ_{V5}) and finite memory time (τ_{V5}).

6.2 Coarse-graining V5 to memory-kernel diffusion

The same substrate-graph averaging applied to V5 yields a **memory-kernel diffusion operator** in the continuum:

$$\hat{D}_{\text{cont}}^{V5} \psi(\bar{x}, \bar{t}) = \int_0^\infty d\tau \mathcal{M}(\tau) \left[D_{V5} \nabla^2 \psi(\bar{x}, \bar{t} - \tau) - \psi(\bar{x}, \bar{t} - \tau)/\tau_{V5} \right]$$

where $\mathcal{M}(\tau)$ is the **memory kernel** inherited from V5's temporal envelope, D_{V5} is the V5 diffusion constant, and τ_{V5} is the V5 memory time.

6.3 Markovian limit

In the **Markovian limit** $\tau_{V5} \rightarrow 0$, the memory kernel becomes $\mathcal{M}(\tau) \rightarrow \delta(\tau)$, and the operator reduces to a Markovian diffusion + decay form:

$$\hat{D}_{\text{cont}}^{V5, \text{Markov}} \approx D_{V5} \nabla^2 - 1/\tau_{V5}.$$

This is the substrate-level structural origin of standard Markovian diffusion + exponential relaxation in soft-matter and condensed-matter contexts.

6.4 Non-Markovian regime

In the **non-Markovian regime**, the full memory kernel $\mathcal{M}(\tau)$ contributes. The resulting continuum equation is a **generalized diffusion equation with memory**:

$$\partial_t \psi(\bar{x}, t) = \int_0^t \mathcal{M}(t - t') D_{V5} \nabla^2 \psi(\bar{x}, t') dt' - \psi / \tau_{V5}.$$

This is the substrate-level structural origin of non-Markovian behavior in soft-matter (Maxwell viscoelastic response with frequency-dependent moduli) and BH-radiation (non-thermal corrections to Hawking spectrum).

7 Cross-Domain Instantiations

7.1 Soft-matter Maxwell relaxation — form derived, value empirically matched

Form (derived). DCGT applied to V5 in the soft-matter Markovian limit produces the **exponential-decay form** of the Maxwell stress-relaxation modulus:

$$G(t) = G_0 e^{-t/\tau_{V5}^{\text{soft}}}.$$

Numerical value (empirically matched, not derived). In the soft-matter regime, τ_{V5}^{soft} is **identified with the empirically-measured Maxwell relaxation time** $\tau_{\text{Maxwell}}^{\text{empirical}}$ via direct empirical matching.

Krieger–Dougherty viscosity and Cross / Carreau / Bird–Carreau constitutive laws emerge as DCGT continuum limits in different soft-matter regimes, with each regime’s specific numerical coefficients similarly inherited via empirical matching.

7.2 QFT propagator hierarchy

In the QFT regime, V1/V5 coarse-graining produces standard QFT propagators:

- **Retarded propagator** G_R : V1 retarded support coarse-grained to continuum retarded propagator.
- **Feynman propagator** G_F : Wick rotation of V1 + V5 in the Euclidean continuation.
- **Advanced propagator** G_A : from G_R via mathematical compositional closure; not directly substrate-operational but mathematically derivable.

V1 finite-width supplies the substrate-level UV cutoff $\omega_c = c/\ell_{\text{ED}}$ that becomes the standard QFT regulator (Pauli–Villars, dimensional, lattice) in the continuum limit.

7.3 Lieb–Robinson velocity bound — form by dimensional analogy

A→dimensional-analogy (not derivation). In the condensed-matter regime, DCGT applied to V5 in the Markovian limit suggests, **by dimensional analysis matching to the standard Lieb–Robinson theorem**, a velocity bound for entanglement-content propagation:

$$v_{LR} \sim c_{V5} \cdot \frac{\ell_{V5}}{\tau_{V5}^{\text{ent}}}.$$

Honest framing: This is **dimensional analogy**, not substrate-level derivation. The standard Lieb–Robinson theorem (1972 and later refinements) is a tight commutator-norm bound from Hamiltonian locality; its derivation requires Hamiltonian-level analysis. ED’s substrate-level v_{LR} form is obtained by matching substrate-level scales to the standard LR-velocity dimensional structure — a defensible identification at the dimensional level, but not a substrate-level proof of the Lieb–Robinson theorem.

7.4 BH curved-spacetime entanglement and KMS

In the BH near-horizon regime, V_5 acquires imaginary-time periodicity (Paper_047 §3). DCGT applied to V_5 in the Euclidean continuation produces the substrate-level KMS condition:

$$\langle \phi(t_A)\phi(t_B) \rangle_{V_5}^{\text{Eucl}} = \langle \phi(t_A + i\beta_H)\phi(t_B) \rangle_{V_5}^{\text{Eucl}}, \quad \beta_H = 2\pi/\kappa,$$

giving thermal Planck spectrum at $T_H = \kappa/(2\pi)$ (Paper_047 §4).

7.5 The procedural unity

The four instantiations are not four independent mechanisms — they are **one substrate→continuum procedure (DCGT) applied to one substrate primitive (V5)** across four regimes that differ in regime-specific τ_{V_5} , ℓ_{V_5} , and substrate-source content. The procedural unity is the cross-domain coherence claim of DCGT.

8 DCGT Breakdown Conditions

8.1 Strong-gradient regimes

In regimes where $L_{\text{flow}} \rightarrow R_{\text{cg}}$ (gradient scale approaches coarse-graining scale), DCGT breaks down: the scale separation $R_{\text{cg}} \ll L_{\text{flow}}$ fails. Examples: shock waves; near-singularity content in BH evaporation late-stage; strong-coupling soft-matter (gels, glasses) where collective-mode wavelength approaches ℓ_{V_5} .

8.2 Near-substrate-scale regimes

In regimes where $R_{\text{cg}} \rightarrow \ell_{\text{ED}}$ (coarse-graining scale approaches substrate scale), DCGT cannot resolve substrate-graph microstructure cleanly. Substrate-level effects (kernel finite-width, V_5 memory, commitment-event statistics) become visible at coarse-grained level. Examples: trans-Planckian regime in Hawking spectrum; sub-Planckian length-scale probes; near-Planck-mass BH evaporation late-stage.

8.3 Near-singularity regimes

In regimes where substrate-level stability-landscape gradients (P12) become large enough to trigger substrate-level commitment-event proliferation, DCGT’s averaging breaks down because commitment events at substrate-scale are not coarse-grain-able. Examples: quantum-measurement events at substrate-graph scale; near-horizon information-blocking transitions.

8.4 Strong-curvature regimes (Arc ED-10 territory)

DCGT operates in flat-spacetime (or approximately-flat) regime. Strong-curvature regimes (near singularities, near horizons at substrate scale) require the Arc ED-10 curvature-emergence framework, not DCGT directly.

9 Falsification Criteria

9.1 Continuum equations fail to match substrate-level predictions + procedural-unity failure

Falsifier: Empirical demonstration that continuum-level equations derived via DCGT (Maxwell relaxation, Newtonian potential, Hawking spectrum, Lieb–Robinson velocity) fail to match substrate-level predictions within the hydrodynamic-window scale separation — systematic

discrepancies that cannot be absorbed into value-layer coefficient adjustment — would falsify DCGT. Equivalently: empirical demonstration that the four cross-domain instantiations require *different* substrate→continuum procedures rather than one DCGT applied across regimes would falsify the procedural-unity claim.

9.2 Hydrodynamic-window failure or breakdown-pattern violation

Falsifier: Empirical demonstration that DCGT breaks down in a regime that should satisfy $\ell_{\text{ED}} \ll R_{\text{cg}} \ll L_{\text{flow}}$, or that breakdown in strong-gradient regimes follows a pattern inconsistent with substrate-graph commitment-event proliferation, would falsify the scale-separation regime condition + the breakdown taxonomy jointly.

9.3 Diffusion-form failure

Falsifier: Empirical demonstration that V1 coarse-graining does not produce a diffusion-type operator in the continuum limit — e.g., produces a non-local integral operator at scales $R_{\text{cg}} \gg \ell_{\text{ED}}$ — would falsify the diffusion-form derivation.

9.4 Memory-kernel structure failure

Falsifier: Empirical demonstration that V5 coarse-graining does not produce a memory-kernel operator with structure inherited from V5 envelope — e.g., that soft-matter Maxwell relaxation does not follow the predicted memory-kernel form — would falsify the V5 coarse-graining derivation.

10 Position Statement

This paper sits within the substrate-ontology research-program lineage (’t Hooft cellular-automaton; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

The **Diffusion Coarse-Graining Theorem (DCGT)** is the substrate→continuum bridge in the ED Generative Primitives System. Given the postulated primitives + V1 inheritance + V5 dependence + the hydrodynamic-window scale separation $\ell_{\text{ED}} \ll R_{\text{cg}} \ll L_{\text{flow}}$ (regime condition), substrate-level kernel-mediated content coarse-grains to continuum diffusion-type / propagator-type / constitutive-law equations: V1 → diffusion operator; V5 → memory-kernel diffusion; cross-domain instantiations across soft-matter (Maxwell relaxation), QFT (propagator hierarchy), condensed-matter (Lieb–Robinson velocity), and BH (KMS-thermal structure). The procedural unity — one substrate→continuum procedure applied across multiple regimes — is the cross-domain coherence claim of DCGT. Breakdown conditions specify where substrate-level mechanism is *not* captured by continuum description; these breakdowns are predictions, not failures, and provide falsifiability handles distinguishing ED from continuum-only frameworks.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) within hydrodynamic-window scale-separation regime per Paper_095 §3.3. Form derived via V1/V5 coarse-graining + diffusion-form emergence + memory-kernel structure + cross-domain instantiations; values inherited (Maxwell τ , transport coefficients, regime-specific parameters); regime explicit (A→regime hydrodynamic-window).

What this paper claims: given the postulated primitives + V1/V5 + hydrodynamic-window regime, DCGT produces continuum diffusion-type / propagator-type equations as substrate-level structural consequences; the four cross-domain instantiations are one substrate→continuum procedure applied to one substrate primitive (V5) across regimes; DCGT is load-bearing for

Papers_027 (Newton), 047 (Hawking), 090 (V5 cross-domain), 056 (Class-A coarse-grained matching), and the entire Wave-3 corpus inheriting at M3-equivalent level.

What this paper does *not* claim: the substrate primitives are not derived (Paper_087); DCGT is not the only substrate→continuum bridge; continuum-level transport coefficients are not derived but inherited; DCGT does not apply outside the hydrodynamic window; DCGT does not derive Einstein equations or standard QFT renormalization; the Lieb–Robinson identification is A→position dimensional-analogy framing, not substrate-graph derivation.

The empirical case rests on cross-domain reach: DCGT is load-bearing across multiple arcs; falsification in any arc traces back to DCGT or to the arc-specific application.